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Review article

On biological and artificial consciousness: A case for biological computationalism

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ABSTRACT

The rapid advances in the capabilities of Large Language Models (LLMs) have galvanised public and scientific debates over whether artificial systems might one day be conscious. Prevailing optimism is often grounded in computational functionalism: the assumption that consciousness is determined solely by the right pattern of information processing, independent of the physical substrate. Opposing this, biological naturalism insists that conscious experience is fundamentally dependent on the concrete physical processes of living systems. Despite the centrality of these positions to the artificial consciousness debate, there is currently no coherent framework that explains how biological computation differs from digital computation, and why this difference might matter for consciousness. Here, we argue that the absence of consciousness in artificial systems is not merely due to missing functional organisation but reflects a deeper divide between digital and biological modes of computation and the dynamico-structural dependencies of living organisms. Specifically, we propose that biological systems support conscious processing because they *(i)* instantiate *scale-inseparable*, substrate-dependent multiscale processing as a metabolic optimisation strategy, and *(ii)* alongside discrete computations, they perform continuous-valued computations due to the very nature of the fluidic substrate from which they are composed. These features – *scale inseparability* and *hybrid computations* – are not peripheral, but essential to the brain's mode of computation. In light of these differences, we outline the foundational principles of a biological theory of computation and explain why current artificial intelligence systems are unlikely to replicate conscious processing as it arises in biology.

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1. Introduction

Since Descartes watched a 17th-century hydraulic automaton move with an uncanny human likeness (Makari, 2015), science has grappled with the haunting proximity between mechanism and mind. From these early machines, crafted from pistons and springs, to today's digital computers built of silicon and code, the question remains: what separates lived experience from clever, automated execution?

Descartes, of course, framed this in terms of dualism – *the soul and the machine*. However, the same tension persists in modern debates. As the fluency of Large Language Models (LLMs) fuels speculation about artificial consciousness, the absence of agreed-upon criteria for subjective experience becomes increasingly problematic. This lack of consensus is evident in a review by Seth and Bayne (2022), who surveyed 22 leading theories of consciousness and found little overlap in their assumptions.

Because there is little consensus among consciousness scientists, it is understandable that discussions about artificial consciousness have expanded in both public discourse and related scientific fields as synthetic systems (such as LLMs) have progressed (Graziano, 2017; Butlin et al., 2023; Findlay et al., 2024). For theories advocating for the immediate possibility of developing synthetic systems with atoms of subjective experience, what is often assumed – even tacitly endorsed in contemporary AI research – is a form of *computational functionalism*. That is, consciousness arises wherever the right kind of information processing occurs, regardless of the physical substrate (Butlin et al., 2023). If this is true, then neurobiology, with its morphological and dynamical constraints, is secondary. Only the abstract 'computations' performed by the physical substrate matter.

Opposing this is *biological naturalism*, which asserts that subjective experience depends critically on the physical and biological processes by which it is instantiated (Searle, 2017). From this perspective, it is commonly prescribed that computations necessary for brain function, and more importantly, consciousness, are crucially dependent on the materiality of the system (Cao, 2022). According to this view, consciousness is not just a matter of abstract computation performed on an algorithmic layer, but depends precisely on the kind of system that computes: its morphology, energy, and functional organisation. According to this account, only living systems, and perhaps only nervous systems, support subjective consciousness (Seth, 2024; Aru et al., 2023). What remains largely underdeveloped is an account of what is actually implicated in biological computation, how it works, why it is different from digital, abstract computation, and why it might matter for consciousness.

This paper confronts that gap. From amongst a larger pull of differences, we argue that current digital systems fundamentally diverge from evolved, biological systems in two key ways that are core to computation in biological systems, and that these divergences may help explain why artificial consciousness, with AI's current trajectory, would remain unrealised: not because *synthetic* consciousness is ruled out, but because achieving it would require radically different physical architectures capable of reproducing the substrate-specific computations characteristic of biological brains.

To this end, we propose that biological brains exploit an interdependent multiscale organisation of processes, whereby computations at any one scale are not separable from each other. These scales could be structurally defined scales; molecular, synaptic, columnar, or population levels (Munn et al., 2024), or purely functional and dynamical scales of information dynamics (Milinkovic et al., 2025b), where their scale-interdependence was shown to be stronger than in states of abolished consciousness (Milinkovic et al., 2025b). This *scale inseparability* suggests that there is no single privileged scale of dynamical processes that underpin consciousness, unlike digitally engineered systems which are inherently *scale-separable* and hierarchically decomposable systems (Rosas et al., 2024). Here, we develop this argument further to suggest that conscious processing, *in general*, could be dependent on this scale-integrated dynamical structure. We argue that the brain realises such a scale-integrated functional organisation precisely as a metabolic optimisation strategy (Deacon, 2011), to overcome severe energy scarcity.

Artificial Neural Networks (ANNs) which run on von Neumann hardware, are unconstrained by biological energy scarcity. In fact, their progress and success in performing more complex functions hinges on *more*: more compute, more parameters, more energy. The brain, by contrast, has succeeded through *less*: by refining how structure and function co-evolve under constraints. In turn, accreting a *dynamico-structural* dependency, that digital, solid-state systems just do not possess.

Alongside scale-integration, our second consideration is that, concomitantly with discrete-valued operations such as neuronal action potentials, the brain also realises its functional capabilities through continuous-valued signals through field propagations. These signals occur across multiple structural and functional scales, including non-graded potentials, electric fields, and their associated phenomena such as ephaptic coupling and oscillatory behaviour. Research indicates that large-scale ionic gradients, a continuum property, also provide a mechanism for macroscopic control over the excitability of localised neural populations (Halnes et al., 2016; Rasmussen et al., 2020; Chesebro et al., 2023). With this in mind, we expand on the notion of computability and propose how formal theories, inspired by brain mechanisms, could guide strategies for incorporating continuous-valued, field-like computations into synthetic systems.

Thus, we propose that a theory of *biological computation*, one that could meaningfully inform discussions about the possibility of subjective consciousness in *other things*, must account for the two biological features of operation currently missing in artificial systems. First, the inherent scale-inseparability of multiscale biological processes, which operate as a metabolically optimised strategy wherein computations at one scale are not cleanly separable from computations at other scales. Second, the continuous-valued electrochemical dynamics that endow neural tissue with a hybrid discrete-continuous mode of computation.

Importantly, we do not claim to be explaining consciousness itself, nor to be specifying *why* consciousness exclusively arises from these biological processes. Rather, we aim to offer an alternative way of thinking about consciousness, one grounded in a more accurate understanding of the operations and computational principles exhibited by evolved biological systems. We argue that these features are not peripheral but essential for framing how consciousness arises in the brain, and why current digital architectures are unlikely to reproduce it. To highlight this contrast, we next examine the defining features of digital systems and the computations they perform.

2. Digital computation: A tale of separability

2.1. Here, there, and not everywhere

In computer science, computation is defined as the rule-governed manipulation of *discrete* symbols on a von Neumann machine (Sipser, 1996; Wolfram, 1984). At the hardware level (see Fig. 1a), von Neumann machine are designed around three physically distinct units

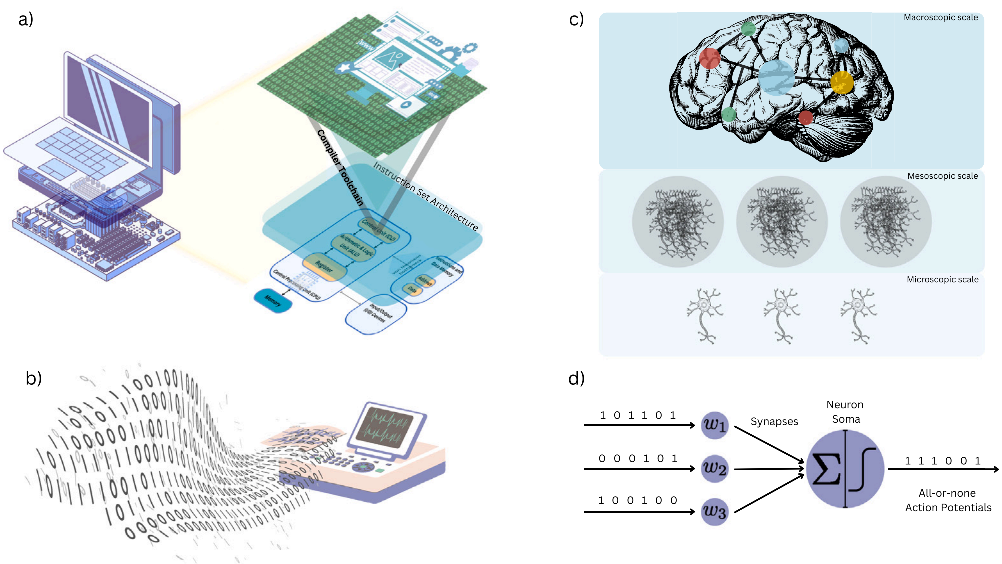

Fig. 1. The persistence of the digital-and-discrete computation metaphor in neuroscience. (a) Classical digital machines are organised around separable hardware components (memory, arithmetic–logic unit, control unit) linked via an instruction set architecture and compiler toolchain. (b) This architecture embodies the assumption that computation is defined over discrete-valued systems, an assumption that has carried over into neuroscience. (c) Neural processes are commonly framed through three arbitrarily separated scales (microscopic, mesoscopic, and macroscopic) whose constituent units are treated as functionally homogeneous. (d) The integrate-and-fire neuron exemplifies this inheritance: conceived as a device that converts discrete inputs into discrete outputs, it mirrors the abstraction of symbol manipulation in digital systems. This illustrates the tenacity of computational analogies that frame the brain in digital terms, i.e., an assumption this paper seeks to critically examine and move beyond.

connected by a bus. Similar to buses in traffic, this is a communication system that transfers data between these 3 physically isolated units, which include: (i) the *memory* unit that passively stores discrete symbols, (ii) the *arithmetic-logic unit* (ALU) that manipulates those same discrete symbols, and (iii) the *control unit* that fetches an instruction, decodes it, and dispatches it back to the ALU. In addition, program instructions and stored data share a uniform address space physically separated from the ALU which constrains the computational performance. Formally, this separation induces what is called a *von Neumann bottleneck* and is a constraint on performance which is tolerated because it enforces a clean separation between the place where symbols are stored and the mechanism that manipulates them (Backus, 1978).

Modern digital computers extend this separability of components to a separability of scales. The processor's *Instruction Set Architecture* abstracts away from the underlying digital circuits and transistors, enabling binary-code compatibility across fabrication technologies (Patterson and Hennessy, 2016). Compilers then translate high-level programs into *intermediate representations* and finally into machine code, preserving functional equivalence whilst insulating algorithms from their physical substrate. This forms a *hardware abstract layer* in which algorithms and their physical realisers are designed to be independent (Patterson and Hennessy, 2016). Formalised as the *reduced instruction set computer* design, abbreviated RISC, it is the archetype of software-hardware separability. This clean separability between *symbol* and *substrate* is a deliberate design principle that runs from Turing's *stored-program* idea (Turing, 2004) to modern ANNs.

ANNs also inherit this separability. At the hardware level, GPUs shuttle tensors between DRAM and SRAM via *direct memory access*. repeating this cycle layer by layer. At the software level, frameworks such as PyTorch and TensorFlow construct static computational graphs (the network operations) that remain distinct from the data flowing through them. Further, training of the ANNs alternates between a forward pass (computing activations) and a backward pass (computing gradients), while the weights themselves remain as static data in memory (Ketkar et al., 2021). Crucially, the learning rule (back-propagation) does not depend on the specific data points or where the weights are stored. This fungibility enables data-parallel training to scale across thousands of GPUs (Hecht-Nielsen, 1992).

These engineered separations yield modularity, fault isolation, and portability. But they also instantiate a computational ontology in which what is computed (symbolic manipulation) is insulated from how it is physically realised (Fig. 1a–b). Separations between symbol and substrate, software and hardware, functional and realisation, are also the conceptual foundations on which computational functionalism rests: that algorithmic and computational levels, and by extension consciousness, are substrate-independent, i.e., separable from the implementation level (Marr, 2010). It is therefore necessary to make explicit the commitments of computational functionalism and the ways in which it inherits these engineering assumptions.

2.2. Computational functionalism: Its commitments and limits

Computational functionalism remains the dominant theoretical stance in contemporary discourse on artificial consciousness (Butlin et al., 2023; VanRullen and Kanai, 2021; Seth, 2024). At its core, it asserts that mental states and conscious experience supervene solely on the right kind of computational and algorithmic organisation, irrespective of the substrate in which that organisation is instantiated (Putnam et al., 1967; Lewis, 1972; Chalmers, 1996). This view inherits, often implicitly, the engineered separability outlined above. That is; consciousness is wholly determined at the computational and algorithmic levels: if the algorithmic structure is preserved, consciousness should in principle follow, whether in neurons, silicon, or any other medium.

Under this assumption, the theoretical question naturally becomes: *Which computational properties are necessary and sufficient for consciousness?* Evidently, this question divorces computation from its physical realisation entirely: leaving no space for the effect of the physical substrate on the computations performed.

Increasingly, however, work across philosophy, neuroscience and the physics of computation is revealing the limits of this view. Thagard (2022) emphasises that substrate-independence neglects the energetic and thermodynamic realities of information processing: computation is never free-floating but requires concrete physical processes with non-trivial metabolic costs (Parrondo et al., 2015). Piccinini (2020) further shows that computational functionalism conflates two claims: that consciousness is functional, and that it is computational in the classical digital sense. Under his view, even if consciousness is functional, it does not follow that it is Turing-computable or substrate-independent.

Shagrir further (Shagrir, 2005) notes that early historical arguments for multiple realisability – the notion that a function can be realised on a multiplicity of different physical substrates – relied on oversimplified assumptions about both computation and neurobiology, while more recent analyses question whether mental states are multiply realisable in the strong sense required (Cao, 2022). Further, Kleiner (2024) proposes that if consciousness is a computation, it cannot be a classical Turing computation at all; it would instead require a more physically grounded form of mortal computation (Kleiner, 2024) defined by finite, substrate-specific operations rather than idealised, potentially unbounded Turing-style procedures (Hinton, 2022). In the general sense, mortal computation refers to computation whereby the machine's mortality – its requirement to halt – extends computational power beyond classical Turing machines. Lastly, Findlay et al. (2024) demonstrate that functional equivalence does not entail experiential equivalence, showing that digital simulations can preserve functional structure while losing causal structures thought to be essential for consciousness.

These critiques show that computational functionalism relies on an impoverished conception of computation derived from von Neumann digital computers: machines that cleanly separate scales, discretise dynamics, and treat physical time and computational time separately (Fig. 1).

Biological tissue, by contrast, implements continuous electrochemical processes directly in the substrate: the physical dynamics are the computation. This collapses the usual algorithm-implementation distinction and exposes what substrate-independent digital architectures fail to capture.

Framed this way, the central question becomes: *Which physical properties of biological substrates are necessary for the computations underlying consciousness?* Addressing this requires us to move from computability as discrete-state symbol manipulation and its implementation on contemporary digital infrastructures, to computation that takes its physical implementation as foundational and includes physical time continuous dynamics. As pioneers of neuromorphic engineering have emphasised (Mead, 2002, 2023), the substrate is not merely a carrier of the algorithm—the substrate *is* the algorithm (Indiveri, 2025).

This motivates our next step. Computational functionalism is anchored in the classical Turing-Church model of discrete symbolic computation implemented in von Neumann machines. Modern ANNs inherit this blueprint despite Turing's reminder that digital computers need not be discrete-state machines in any ontological sense (Turing, 2004). To understand the scope and limits of this framework, and why it may be inadequate for modelling biological conscious processes, we now turn to the formal foundations of computability itself.

2.3. On computability in formal systems

To proceed, we must examine the foundations of computability: that is, which functions are computable in principle, and which are provably not? Addressing this question is essential for assessing which classes of computation; whether mortal (Hinton, 2022), quantum (Penrose, 2016; Hameroff and Penrose, 2014), neuromorphic (Ulhaq, 2024), or something yet undescribed (Aru et al., 2023) might be uniquely implemented in evolved, biological systems, and why.

In modern terms, a function is computable if there exists a finite, mechanical procedure (formally, a Turing machine) that can output the function's value for any valid input in finite time (Turing et al., 1936). Computability, in this sense, was first formalised by Turing in response to Hilbert's *Entscheidungsproblem* (Hilbert, 2019): determining when a number-theoretic function, grounded in the axioms of Peano arithmetic, is computable (Turing et al., 1936; Cooper, 2017). Turing formalised this notion through the *universal Turing machine*, a purely theoretical device which could simulate any effectively calculable (or computable) function given unbounded resources. This formulation presaged the von Neumann architectures; modern digital computers are direct inheritors of this universal machine, reduced to operations on binary logic gates and discrete symbol manipulation.

In discrete natural number arithmetic, *in*computability emerges at the heart of these formal systems (Cooper, 2009). Here, incomputability is defined as the inversion of computability; i.e., as the condition where no such algorithmic procedure exists for computing the output value of a function. The paradoxical nature of systems in which incomputability is realised is well encapsulated in Kurt Gödel's work. Gödel's incompleteness theorem (first published in 1931) guarantees that any sufficiently powerful formal system that can be expressed arithmetically through natural numbers (e.g., Peano arithmetic) is *incomplete* (Gödel, 1992). That is, there are true statements (outputs of a function) about natural numbers that cannot be proved (another output of true or false values) within that same system. A hidden feature often misrepresented in Gödel's proof is the following: the valuation of whether a statement is true or false is itself a function that exists at a different level than the statements being evaluated. This reveals an implicit hierarchical structure in formal systems: to determine truth at one level, one must appeal to a higher level.

Initially, Turing proposed a partial solution by introducing *oracle machines*, whereby a Turing machine can query an 'oracle' that instantly provides answers to its queries (Turing, 1939). While this implicitly assumes another computational system that operates above, or external to, the querying machine, it does not provide a systematic theory of how these levels relate or why they are necessary. Alfred Tarski's *undefinability theorem* makes this necessity explicit and provides the formal reason for hierarchical systems: truth for arithmetic systems cannot be defined within arithmetic itself (Tarski, 1933). One requires a meta-level: a second formal system capable of expressing truths about the first. This meta-level, in turn, harbours its own incompletenesses requiring yet another level above it. This process iterates infinitely, forming what are known as *Tarski hierarchies* (Halbach, 1995). Each level in this hierarchy can resolve some truths inaccessible to lower levels, generating a multiscale structure where definability are intertwined across levels. Crucially, information at the higher level is essential for determining definability and truth at the lower level, and visa versa, information at lower levels are necessary for the higher level to have anything to evaluate, and therefore the levels cannot be cleanly separated. This reveals a formal notion of hierarchical organisation in logical systems: to settle truth at one level often requires moving to a meta-level with greater expressive power.

Rubel (1993) extended this insight into the computational domain, showing that adding successively richer sets of computational primitives (effectively Turing's oracles for specific functions), analogue computers capable of continuous computations – not just discrete symbol manipulation – can approximate increasingly broader classes of real-valued functions. His *extended analogue computer* operates through a hierarchy of levels, where each level includes all primitives from lower levels plus additional functions. This shows that increasing the number of accessible computational scales (moving up the hierarchy) strictly increases what can be computed. However, Rubel's levels remain separable: each level is defined independently, and higher levels provide fixed oracle-like functions that lower levels can query without feedback. This unidirectional structure differs fundamentally from biological systems, where we observe bidirectional coupling: lower scales *endogenously generate* higher scales, and higher scales simultaneously constrain lower scales through continuous, co-determining dynamics.

With respect to consciousness, Roger Penrose famously leveraged Gödel's theorem in *The Emperor's New Mind* (Penrose, 2016), arguing that human consciousness and thought cannot be purely algorithmic (note that his argument is specifically isolated to discrete systems). His reasoning is Gödelian: if human mathematicians were equivalent to discrete formal systems, then there would exist true mathematical statements they could not recognise; yet humans demonstrably can see the truth of certain Gödelian statements (at least in principle). Penrose thus infers that consciousness involves non-computable processes, potentially grounded in unknown physical laws, possibly linked to quantum gravity (as later developed in his Orch-OR hypothesis with Hameroff and Penrose, 2014).

This claim remains controversial because it conflates human reasoning (and consciousness) with formal provability in arithmetic discrete systems. Further, Penrose's argument neglects two crucial considerations. First, it ignores the Tarski hierarchies and the possibility that consciousness might involve scale-integrated computation across multiple levels rather than being confined to a single formal system. Second, and more fundamentally, it overlooks continuous-valued and continuous-time systems. We believe these considerations constitute a fundamental and important caveat in formal systems comparable to Gödel's own famed proof. Thus in our view, living systems might exploit physical dynamics that are not naturally captured by single-level, discrete formal systems, even if they remain Turing-computable in principle.

Further, David Deutsch's work on the *Church-Turing-Deutsch* principle (Deutsch, 1985) generalised Turing computability to quantum systems, showing that even quantum computers, despite their exponential speed-ups, remain bound by the same class of computable functions as classical digital machines. Quantum algorithms do not circumvent Gödelian limits: they do not compute uncomputable functions, nor do they evade Tarski's undefinability. Thus, even quantum computation is still, in principle as well as practice, Turing-equivalent. This result is crucial: simply invoking quantum physics does not break through the limits of computability revealed by Gödel (Tarski et al., 1953). If living systems do increase classical (Turing) computation (without recourse to arguments for non-Turing computations), it must be due to something beyond quantum effects. Perhaps their continuous, field-like dynamics or multiscale organisation where scales are inseparably coupled.

Turing himself also recognised broader notions of computation, including those defined over real numbers, i.e., continuous computations (Turing, 1939). Subsequent theoretical developments in real-valued computation have advanced considerably, yet have rarely been carried over into neuroscientific discourse (Braverman and Cook, 2006). From early beginnings, Claude Shannon, the godfather of modern information theory, demonstrated that differential analysers – analogue computers built from integrators and polynomial operations – can generate functions satisfying algebraic differential equations (the so-called *non-hypertranscendental functions*) (Shannon, 1941), a result which was later refined for General Purpose Analogue Computers (Campagnolo et al., 2000). More contemporaneously Blum et al. (1989) developed a theory of computation over rings; a mathematical structure that includes the real number line. However, their model remains fundamentally discrete-step with real-valued registers rather than capturing continuous evolution directly (Blum, 2004; Blum et al., 2012; Braverman and Cook, 2006). Lastly, MacLennan pioneered continuous spatial automata and field computation, emphasising physical realisability and natural computation in a 3-dimensional space (MacLennan, 2004, 2012), yet lacked formal characterisation of computability. These developments show promise in the intriguing benefit of continuous computations to complete discrete systems. Despite these advances, none of these frameworks simultaneously capture hybrid discrete-continuous computation, scale-inseparable multiscale dynamics, and substrate-dependent physical realisation, all realised in a stochastic universe; features we will argue are essential for biological computations that might underlie consciousness.

The fascination with digital (discrete-state) machines and their spectacular success owes much to von Neumann architectures, which meant that continuous computation was largely overlooked, even though early work had already hinted at its very different properties. One of these is a remarkable and often overlooked result from Tarski. It develops an exposition that the first-order theory of the real numbers with addition, multiplication, and order – the so-called real closed fields – is both; complete and decidable (Tarski, 1951). Using quantifier elimination, every polynomial statement about real numbers can be algorithmically transformed into an equivalent quantifier-free statement. In striking contrast to the incompleteness of natural number arithmetic, real number arithmetic admits a full decision procedure: every statement in this language is, *in principle*, computable.

This inversion is profound and has suggestive implication for how we model biological computation. Whilst discrete natural number arithmetic (the foundation of digital computers) is fundamentally marked by incompleteness and incomputability, continuous arithmetic over the real numbers admits a full decision procedure *at the level of the formal system*.

As we will see, neural processes unfold within a continuous fluidic substrate whose dynamics are undoubtedly governed by real-valued quantities – voltages, concentrations, field strengths – that all show properties of continuum dynamics. While biological systems might not implement Tarski's decision procedure exactly, the contrast is nonetheless instructive. Digital computers must represent continuous quantities through discrete encodings and simulate differential dynamics symbolically. Biological systems, by contrast, implement these dynamics directly in physics: ion flows, membrane potentials, receptor kinetics, dendritic integration, and extracellular fields evolve continuously without symbolic mediation. We will describe some of the properties in the upcoming section. This does not grant biological systems hypercomputational (non-Turing) power, but it does endow them with a computational mode inaccessible to symbol-manipulating machines: one in which the substrate inseparably realises the algorithm. In this sense, continuous dynamics may offer functional advantages that cannot be recovered by discrete approximations, regardless of how fine they are.

As we will see, the scale-integrated nature of neural dynamics, where microscale molecular events, mesoscale population dynamics, and macroscale brain-wide activity are inseparably coupled, may be analogous to the hierarchical structure highlighted by Tarski hierarchies: just as higher logical levels are required to resolve incompleteness at lower levels, higher biological scales may be required to resolve computational challenges intractable at lower scales. However, unlike Rubel's separable oracle hierarchy or Tarski's unidirectional meta-levels, biological scale-integration involves bidirectional coupling where scales mutually and endogenously generate and constrain one another. In the following sections, we outline how this continuous-discrete hybrid structuring, embedded in metabolically-constrained, stochastic, scale-integrated dynamics, may constitute a fundamentally different kind of computation: one capable of supporting conscious processing.

3. Biological computation: Substrate, scale, and hybrid dynamics

This section examines how biological computation, characterised by hybrid processes coupled across inseparable scales, is realised in neural systems. Unlike engineered systems built from isolated modules and isolated scales computing discrete symbols, biological organisms, and the brain in particular, are evolved systems: constructed over time through a messy process of *accretion* and reconfiguration rather than by a clean top-down engineering design (Feinberg and Mallatt, 2016). Accretion refers to the evolutionary layering of new structures and functions atop existing ones, whereby newer capabilities do not replace but rather build upon and remain dependent on ancestral mechanisms. Contrasting this to modular engineering, where components can be designed independently and then assembled, accretion integrates new and old, creating systems where later-emerging functions are structurally and dynamically inseparable from their evolutionary precursors. Accordingly, this feature of accretion is suggested to be crucial for the evolutionary emergence of consciousness (LeDoux, 2022, 2023).

In light of this, and as much as the computational metaphor has advanced neuroscience (Fig. 1c-d), neural processes are neither neatly modular nor scale-separable. The activity of a single neuron, for instance, is modulated by ionic gradients (concentrations of sodium, potassium, calcium, etc.), neighbouring electric fields (ephaptic coupling), astrocytic control of metabolic expenditure, and local-field entrainment. Another example is found in the overlapping involvement of neural populations in distinct functions (Pessoa, 2023): the brain's reuses or its machinery is the rule, not the exception. Multiscale dynamical interdependencies such as these exemplify that later-emerging whole-brain functions are scaffolded on earlier ones and are continually shaped by their broader dynamical contexts (Deacon, 2011). This perspective of multiscale interdependencies also resonates with early accounts of consciousness as an emergent, macroscale property that supervenes on but is not reducible to local neural interactions, but rather, functions to co-determine the underlying neural interactions (Sperry, 1969).

Notwithstanding the centrality of multiscale interdependencies of biological processes, the traditional computational metaphor has of the brain has nonetheless been pervasive; a notion inherited from traditional computing paradigms and in turn influencing contemporary approaches to AI (illustrated in Fig. 1). This metaphor entails a privileged scale for what counts as the fundamental computational unit of the system. In the computational functionalism view of consciousness (Butlin et al., 2023), it is neurons that are typically cast in this role. Larger scales, such as the mesoscopic scale of neural populations or macroscopic scale of whole-brain dynamics (Fig. 1c), are said to be only qualitative, low-dimensional descriptions of brain function, without their own computational or causal processing. Similarly, smaller scales, intracellular and molecular continuum dynamics, are mostly ignored as enabling conditions.

Yet as we will see, decades of neurobiological research spanning cell morphology to whole-brain dynamics, suggest this view is incomplete. In reality, along with discrete events on single scales, neural processes are equally characterised by scale-integrated and continuous dynamics: continuous molecular processes, graded membrane potentials, oscillatory fields, and neuromodulatory gradients that unfold across space and time. Importantly, these continuous processes are not secondary; they actively shape, and are shaped by, the discrete events they surround. This duality complicates any attempt to locate single fundamental units of computation. If spikes represent discrete events, local field potentials represent the continuous background from which they emerge; both co-determine neural function.

This interdependency of continuous and discrete processes across scales invites us to reconsider the very notion of computation in evolved biological systems and its relation to conscious processing. The evolutionary process of accretion and its result in coupling structural and functional scales provides a steep challenge for any attempt to identify a privileged scale of neurobiological computations that give rise to consciousness. As early as the 1990s, Varela warned of this limitation, urging a rethinking of how computation might be understood in biological systems (Varela, 1997; Bechtel and Bich, 2021). The current advent of AI, its comparison with our own brains, and the growing need for clearly articulated distinctions makes this consideration timely again.

Below, we begin by outlining how the multiscale interdependencies of the brain's dynamical structure emerges as an optimisation strategy imposed by metabolic constraints, giving rise to hybrid computation across inseparable scales: a myriad scaffolding of cell morphologies, field effects, and oscillatory dynamics fundamentally different from engineered systems and offering concrete clues as to how consciousness might arise from biological substrates (Bassett and Gazzaniga, 2011).

3.1. Metabolic constraints as an architectural driver

Biological computation operates under stringent energy limits, and this constraint is not merely a boundary condition but a driving principle of neural organisation. Although the brain represents only 2% of body mass, it consumes about 20% of total metabolic output, most of which supports ion pumps maintaining membrane gradients for synaptic transmission (Levy and Baxter, 1996; Niven and Laughlin, 2008). Rather than succumbing to functional limitations, neural dynamics appear to circumvent energy scarcity through structural and functional innovations that simultaneously reduce cost and enhance computational capacity.

Energy optimisation is visible even in single-channel biophysics. Individual neurons tune their ion-channel kinetics to minimise ATP expenditure, preferring the metabolically cheaper strategy of boosting potassium conductance over packing in more sodium channels when higher firing rates are required (Hasenstaub et al., 2010). But the brain's response to metabolic constraints extends far beyond cellular housekeeping; it fundamentally shapes computational architecture through coarse-graining strategies that integrate information across scales, helping the brain perform ever-more complex functions.

We note that spikes are not the sole carriers of neural information. Across both central and peripheral systems, non-spiking neurons use graded potentials to relay signals, serving as continuously computable intermediaries in sensorimotor loops, particularly at ribbon synapses, where continuous transmitter release enables high-fidelity, low-cost transmission (Magistretti and Allaman, 2015). This is absolutely crucial for information transfer; intracortical recordings suggest that such graded neurons can carry up to five times more bits of information per second than their spiking counterparts (Laughlin et al., 1998). Because many incoming spiking neurons converge onto single non-spiking neurons, these findings point to the computational and informational utility of coarse-graining neuronal signals: aggregating discrete spiking inputs into more reliable, continuous macroscopic processes (De Ruyter van Steveninck and Laughlin, 1996). Yet, such coarse-grained continuous dynamics are not independent abstractions but remain fundamentally integrated with their microscopic, discrete origins, a bidirectional coupling where lower scales generate higher scales, and higher scales constrain lower scales. Here we see simultaneous activity of hybrid, discrete-continuous, strategies to carry information and compute: a spectrum of cost-aware processing absent from current ANNs.

This evidence suggests that the brain harnesses the ability to spatially coarse-grain signals, fundamentally transforming how dynamics are organised. This occurs because projecting information through graded potentials rather than spike trains (which are more energetically costly) reduces metabolic cost while extending computational and transmission capabilities.

As argued by Winning and Bechtel (2018), biological systems realise computational power not through fixed, modular architectures but via flexible, temporally extended constraints governing processes across space, time, and energetic domains. In a broader context, these constraints underpin biological autonomy (Moreno and Mossio, 2015), enabling self-organisation and active maintenance. Energy limits do more than constrain performance; they actively shape the repertoire of mechanisms, from ion-channel selection to graded synaptic transmission, that the brain recruits to perform work and stay in energetically favourable states (Friston, 2010). Metabolic constraints are therefore constitutive forces, shaping how neural systems process, transmit, and integrate information across scales (Deacon, 2011), making scale-integration and metabolic embeddedness foundational principles of biological computation.

3.2. Multiscale organisation: Heterarchy not hierarchy

The brain's response to metabolic constraints – that of coarse-graining information across scales and retaining multiscale interdependencies – creates a particular form of organisation fundamentally different from engineered hierarchies. Traditional models depict information flowing up cortical hierarchies (from sensory input to abstract representation) and top-down commands trickling down. However, anatomical connectivity reveals many long-range and parallel connections defying such linear bidirectional models. Hawkins et al. argue that the neocortex is more aptly described as *heterarchical* rather than strictly hierarchical (Hawkins et al., 2025). Individual cortical columns appear to learn sensorimotor relationships independently and in a distributed fashion, reflecting both, hierarchical and heterarchical dynamics.

Heterarchical models are defined by their rejection of the assumption of homogeneity within scales. Neural units at each level exhibit vast heterogeneity in both space and time, an insight philosophers have dubbed rainforest realism (Ross et al., 2007). This view challenges the idea of computational or causal closure at any given scale (Bedau, 1997; Rosas et al., 2024), insisting instead that brain dynamics emerge from continuous interactions across scales and functional domains.

This heterarchical organisation echoes, but also extends, formal structures in logic and computation. Recall Tarski hierarchies: truth at level k cannot be defined within level k but requires meta-level $k+1$. Each level resolves incompleteness of lower levels, generating multi-scale structure where definability and computability intertwine across levels. Rubel's extended analogue computer demonstrated that adding computational levels (oracles for specific functions) strictly increases computational power (Rubel, 1993). However, Rubel's levels remain separable, without a notion of heterogeneity of the computational units at each scale: rather, each are defined independently, with higher levels providing fixed oracle functions that lower levels query unidirectionally without feedback.

Biological scale-integration differs fundamentally: it involves bidirectional coupling where scales mutually generate and constrain one another. Information at higher scales is essential for determining computability at lower scales, yet higher scales are generated by lower scales. This reveals a formal notion of scale-integration that assumes heterarchy over hierarchy: proving statements at one scale requires simultaneous access to information from other scales, much like Tarski hierarchies, but with continuous co-determination rather than unidirectional queries.

Much of the difficulty in appreciating heterarchical dynamics stems from conventional network-theoretic models, which reduce brain processes to schematic feedback loops or pairwise connectivity patterns. These abstractions lend themselves to hierarchical visualisations (as in Fig. 1c) but miss scale-integrated, continuously interacting processes realised in neural tissue as illustrated in Fig. 2.

The contrast highlights an important conceptual caution: importing metaphors from engineered systems (which assume clean hierarchies) can obscure the functional organisation of biological brains. We require frameworks accounting for multiscale integration, dynamico-structural co-determination, and the fluid, heterarchical nature of evolved neural processes.

3.3. Hybrid computation across scales

Having established that metabolic constraints drive scale-integration and that this organisation is heterarchical rather than hierarchical, we now examine exactly how hybrid, discrete-continuous computation is realised across scales: from subcellular biochemistry to network-wide oscillations.

3.3.1. Subcellular: Continuous accumulation drives discrete events

Evidence for diffuse, continuous processes at the subcellular level exerting profound effects on network-level transitions extends our understanding of graded potentials. These processes can be considered emergent temporal coarse-grainings, or slow features, that evolve over longer intervals than fast electrophysiological activity. Thornquist et al. show that Protein Kinase A (PKA) compares internal activity with that of postsynaptic neurons, functioning as a decision process or a continuous evidence accumulation mechanism determining when the network's dynamic goal has been reached (Thornquist et al., 2021). Once this point is achieved, PKA drives a transition to a synchronised burst of calcium influx (Gautham et al., 2024). Although the large-scale calcium surge appears as a discrete-like event, and its physiological relevance must be discretely computable, it depends on continuous accumulation of evidence at the subcellular level to push the system to this precipice. This illustrates cross-scale coupling between long-timescale continuous dynamics and physiologically meaningful discrete events: hallmarks of hybrid computation.

A similar principle is evident in astrocytes. Subcellular calcium waves within these glial cells propagate through threshold-dependent mechanisms: unless a sufficient proportion of the astrocyte's arborisation (around 23%) is activated, the wave fails to spread system-wide (Lines et al., 2024). When the threshold is surpassed, however, a full-scale calcium surge occurs, triggering gliotransmitter release and altering neural excitability. This is another striking example where continuous-to-threshold dynamics enforce system-level discrete transitions.

3.3.2. Cellular: Dendritic computation and morphological complexity

While metabolic constraints and subcellular processes shape the brain's computational organisation, the morphology of individual neurons further exemplifies how evolution exploits scales to extend capacity beyond the simplistic "weighted-sum plus threshold" neuron model. Biological neurons are much more complex than these simplified units used in ANNs.

The classical McCulloch-Pitts model (McCulloch, 1949) treats a neuron as a weighted sum of inputs followed by fixed non-linearity (e.g., tanh, sigmoid as in Fig. 1d, or ReLU). Real neurons, by contrast, possess morphologies dominated by highly branched dendritic trees whose geometry and active properties decisively shape computation.

Where a digital perceptron merely sums synaptic inputs and delivers binary output, dendrites perform distributed, non-linear processing (Yuste, 2011). Voltage-gated ion channels densely expressed in dendritic membranes allow dendrites to integrate inputs actively rather than relay them passively, overturning the assumptions of basic cable theory (Koch, 1984; Bédard and Destexhe, 2013; Larkum, 2013). Active dendritic processing supports phenomena such as back-propagating action potentials and graded depolarisations blurring the line between digital spikes and continuous signals (Migliore et al., 1999; London and Häusser, 2005; Larkum et al., 1999).

NMDA receptors add further non-linearity by generating local dendritic spikes travelling towards and away from the soma against usual information flow (Schiller et al., 2000; Antic et al., 2010). This reverse signalling lets dendrites detect the order and timing of synaptic events, endowing them with inherently non-Markovian computational capabilities (Durstewitz and Gabriel, 2007). It remains a challenge for future technological developments to extend neuromorphic systems with analogous mechanisms capturing these dendritically generated spikes and temporally-selective coarse-graining they provide (London and Häusser, 2005). To our knowledge, this is not currently implemented in any neuromorphic systems (Kudithipudi et al., 2025). Branco et al. showed that simply changing sequence or speed of synaptic activation can switch a neuron's output, underscoring dendrites as powerful spatiotemporal filters (Branco et al., 2010).

A key neglected feature of hybrid computations pre-soma are also seen in dendritic spines functioning as nanoscopic biochemical-and-electrical compartments that sequester, filter, and potentiate individual synaptic inputs before forwarding them to parent dendrites. By consolidating presynaptic information in massively parallel, distributed, timing-dependent fashion, they exemplify hybrid computational processes not easily stratified: an organisational principle missing from ANNs, whose units simply collapse all weighted inputs into single undifferentiated sums (Yuste, 2011).

A contrast with digital computers would be useful here to bolster intuition. While dendrites and spines blend continuous membrane dynamics with discrete spike events, conventional hardware stores information in bits and manipulates it with Boolean logic. Inclusive OR

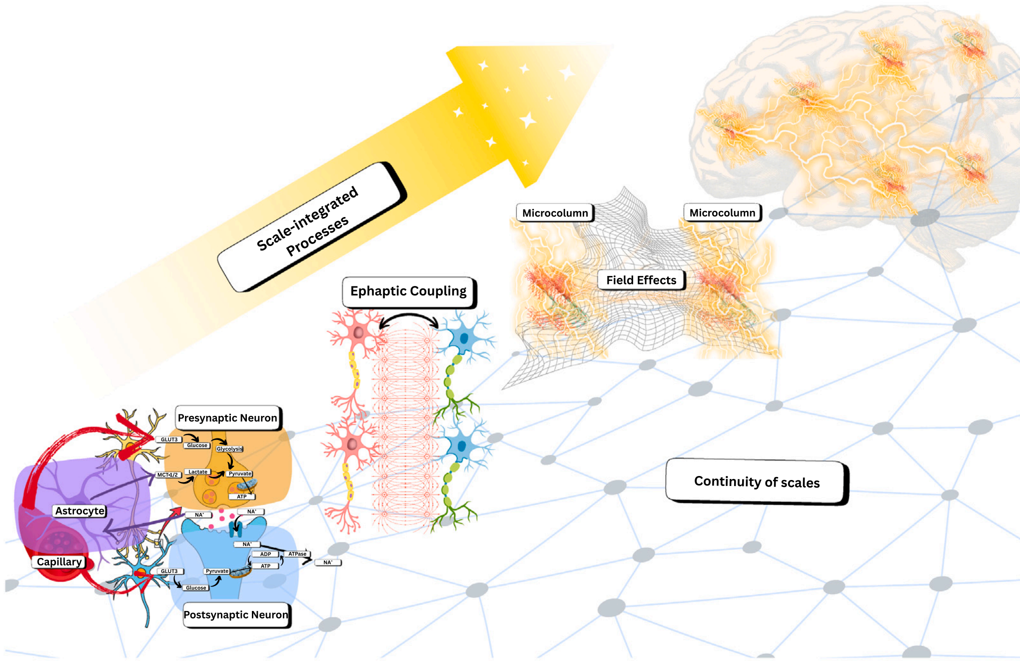

Fig. 2. Continuous, Scale-integrated Biological Computation. Here, we illustrate how neural processes are embedded in continuous, substrate-dependent dynamics that span multiple scales. At the subcellular and cellular level, metabolic constraints shape synaptic transmission: astrocytic regulation of glucose and lactate metabolism influences neuronal energy supply and synaptic efficacy. At the single-cell and population level, ephaptic coupling enables neurons to influence one another through local electric fields beyond synaptic transmission. At the mesoscopic level, microcolumns and local field potentials capture collective dynamics that arise from and constrain single-neuron activity. At the macroscopic scale, these fields extend across regions, coupling activity across distant brain areas. Across all four scales, continuous field-like processes co-determine and integrate emergent dynamics with those of lower levels. In stark contrast to the separable, symbol-based organisation of digital computation, these examples highlight how biological computation is inherently scale-integrated and substrate-dependent.

and AND gates can be reproduced by single-layer perceptrons, but an exclusive-OR (XOR) operation is not linearly separable and demands additional depth in artificial networks (Beniaguev et al., 2021). Recent work shows that a single layer of dendritic computation can solve XOR-like problems. Experiments in layer-2/3 pyramidal neurons reveal that dendritic action potentials embody discrete-continuous processing elicited only within narrow windows of input current ($\approx$ 240 pA) and disappearing when current is too weak ($\approx$ 200 pA) or too strong ($\approx$ 450 pA) (Gidon et al., 2020; Larkum et al., 2022). That stimulus-specific non-linearity enables one biological neuron to match, or even exceed, functional capacity of an eight-layer deep network (Beniaguev et al., 2021).

These dendritic computations diverge sharply from all-or-none events of digital systems. Even when both can, in principle, implement XOR, only the neuron performs the operation while simultaneously temporally coarse-graining incoming signals via back-propagating and locally generated dendritic spikes: behaviours for which current digital systems have no obvious analogue, particularly under the energy limits of dendritic integration. Replicating power of merely a few thousand such neurons would require enormous, densely connected artificial networks. Emerging approaches may narrow the gap, yet efficiency and adaptability of neurobiological processes remain out of reach. Their advantage likely rests on a more general computational framework transcending the classical Church-Turing paradigm on von Neumann hardware, where non-Markovian integration and hybrid processing occur at every scale, with processes at each scale co-determined by processes at other scales. These observations reinforce that the essence of biological computation is not weighted sum plus threshold, but dynamico-structural interdependencies of continuous and discrete processes visible at the scale of individual neurons.

3.3.3. *Populations: Electric fields and non-synaptic communication*

Beyond synapses and dendrites, neurons interact through subtle electric fields (EFs), enabling non-synaptic forms of communication; further blurring discrete modular boundaries. Just as spiking is not the only mode of signalling, synapses are not the only channels of neuronal communication. Neurons also interact through ephaptic coupling: non-synaptic interactions mediated by local EFs modulating the excitability of neighbouring membranes without direct synaptic contact (Jefferys, 1995). By operating as sub-threshold carriers of information local EFs add another layer of spatially extended, energetically efficient communication strategies employed by the brain.

Ephaptic coupling has two principal definitions. First, it describes how neurons or populations communicate in the absence of direct synaptic contact (Anastassiou et al., 2011). Second, it refers to how electric fields“enslave” or constrain underlying neuronal activity, serving as dynamical control parameters (Haken, 1977; Pinotsis et al., 2023; Pinotsis and Miller, 2023). This dual definition highlights continuous operations where ensembles of neurons can exert genuine causal influence beyond classical synaptic interactions (Vyas et al., 2020; Pinotsis et al., 2023).

Experiments demonstrate that weak, *naturalistic endogenous* fields of only a few mV/mm can depolarise neurons, amplify oscillatory frequencies, and synchronise firing-rate activity (Rebollo et al., 2021; Fröhlich and McCormick, 2010) — effects that are not observed at higher field strengths (Fröhlich and McCormick, 2010), and therefore would not appear in exogenous stimulations that might entrain neurons through different mechanisms using higher voltages. This suggests that under natural conditions, relatively weak endogenous fields can significantly shape neuronal dynamics.

Additional evidence comes from biophysical models integrating ephaptic coupling into neural field frameworks built from the first principles of Maxwell’s Equations (Bédard et al., 2004) or empirical data (Rebollo et al., 2021). Some models show that EFs can guide high-dimensional neuronal activity onto stabilised global states, effectively reducing dimensionality (Pinotsis and Miller, 2023). Pinotsis and Miller demonstrate increased information transfer (measured by

Granger causality) when populations are coupled through EFs, mirroring enhanced efficiency of information transmission at coarse-grained levels observed in non-spiking neurons (De Ruyter van Steveninck and Laughlin, 1996). A broader review further introduces the notion of *cytoelectric coupling*, proposing a compelling hypothesis for how EFs may organise and constrain neural activity across multiple scales (Pinotsis et al., 2023).

Like metabolic constraints, ephaptic interactions underscore the importance of multiscale co-determined processing enhancing information transmission. Neural computation spans multiple levels, from ion channels and specialised cell morphologies to electric fields constraining and co-determining spiking network dynamics. This reveals a clear interplay of energy, structure, and function that is fundamentally inseparable.

3.3.4. Temporal: Oscillatory modes as computational syntax

Electric fields, by their nature, are continuous and spatially extended. Unlike discrete signals of synaptic transmission, and hence, can occupy structured states, of which oscillations are among the most prominent. Oscillatory dynamics are ubiquitous in neural systems and have long been proposed as fundamental to cognition and intelligence (Canolty and Knight, 2010; Miller et al., 2024; Singer, 1999; Engel et al., 2001). Yet despite their ubiquity, oscillations have often been dismissed as epiphenomenal: visible in neurophysiological data but not integral to underlying computational processes. This view treats their correlation with cognitive functions as incidental rather than mechanistic.

Recently, however, the causal role of oscillations has gained renewed attention (van Bree et al., 2025). A new dictum is emerging: oscillatory electric fields do not merely accompany neural activity but actively constrain and guide it, shaping the dynamical landscape in which spiking unfolds (Cabral et al., 2022; Pinotsis and Miller, 2023; Bardon et al., 2024; Miller et al., 2024). Importantly, these fields are not best understood as external modulators acting on a separate substrate in which the information for computations is contained; rather, the fields themselves realise oscillatory patterns of organisation that carry information: they encode relational and temporal structure that the neurons interpret, not by reading out symbolic content, but by dynamically entraining their excitability and phase relationships. In this sense, oscillations are information-bearing dynamical modes, not merely carriers of modulation.

Buzsáki and colleagues have suggested that oscillatory activity serves as a form of endogenous coarse-graining: temporal structuring of neural activity that renders spiking events computationally legible to downstream units, whether single neurons or entire ensembles (Buzsáki, 2010; Buzsáki and Watson, 2012). In this view, oscillatory modes constitute a neural syntax: a dynamical scaffold that encodes when neural events matter, what they should be integrated with, and how local activity should be interpreted within broader population structure.

Critically, this informational role is not limited to oscillatory electrical fields generated by neurons. New evidence shows that biological substrate, with its continuum-like physical properties, such as cerebrospinal fluid, also participate in coherent dynamical modes that couple to cortical networks. Campo et al. (2025) demonstrated distinct brain-ventricle coupling modes whose temporal alignment with resting-state networks predicted cognitive ability, challenging neuroncentric views of functional organisation and showing that fluidic-tissue interactions can carry behaviourally relevant information.

These findings underscore that oscillations as continuous, field-like processes are not epiphenomenal by-products of neural activity but constitutive informational-carriers within the brain's energy-aware computational processes.

3.4. Synthesis: Relation between biological computationalism and consciousness

As we see, neurobiological computation does not conform neatly to the digital paradigm. Brains operate at the interface of discrete and continuous domains: spikes (discrete events) ride atop graded membrane potentials, ion flows, and field potentials (continuous processes), which themselves ride atop layer-integrated cortical anatomy (Harris and Shepherd, 2015; Oberlaender et al., 2012). Continuous fields like those of dendritic integration, graded potentials, and electric fields, are central to neural operations yet remain absent from digital systems. Even within neuroscience, the informational roles of such continuous fields often remain peripheral. If aspects of neurobiological computation inhabit this continuous, decidable regime, they may circumvent limits faced by purely digital systems, integrating across scales in ways unavailable to classical von Neumann machines.

Biological computation exploits two complementary mechanisms. First, it leverages multiscale organisation: much like Tarski hierarchies in formal logic, higher organisational scales can adjudicate or resolve limits of computability faced at lower scales, giving rise to emergent, heterarchical dynamics. But unlike Tarski's unidirectional meta-levels or Rubel's separable oracle hierarchy, biological scales are bidirectionally coupled: lower scales endogenously generate higher scales while higher scales simultaneously constrain lower scales through continuous co-determination. Second, biological computation draws on continuous processes: electric fields, ion gradients, oscillatory patterns, and other real-valued dynamics inhabit a domain analogous to real closed fields, where arithmetic is complete and decidable (Tarski, 1951). These features blend the emergence of scales with continuous dynamics into a unified computational strategy.

These capacities may be uniquely suited for conscious experience (Aru et al., 2023). Consciousness exhibits both unity and differentiation (Seth and Bayne, 2022; Albantakis et al., 2023): it presents as a single integrated whole (Metzinger, 2009), yet contains richly differentiated contents. Our aim is not merely to highlight differences between biological and digital computation, but rather, to emphasise that the structural properties of consciousness itself—its unity, differentiation, temporal coherence, and context-sensitivity—appear to require precisely the kind of scale-integrated hybrid dynamics that biological systems uniquely implement.

Theories of consciousness often converge on integration as the hallmark of unified experience, but typically restrict this to a single scale of organisation. This follows from the binding problem: how distributed features are united into one coherent whole of conscious experience (Treisman, 1998; Singer, 2001). Some theories emphasise integration among distributed brain regions (Dehaene et al., 2011), others, amongst the causal connections within local circuits (Albantakis et al., 2023), and others at the cellular level, such as the integration of apical and basal dendritic activity (Aru et al., 2020). Yet in each case, integration is treated as an intra-scale property.

Our account proposes that, in addition to intra-scale integration, inter-scale coupling – that is, integration across spatiotemporal computational scales – may be essential. Just as integration of distributed regions may be required for binding the contents of consciousness, integration of dynamical scales may be necessary for generating the spatially-unified, temporally-coherent flow of conscious life. Recent findings support this view, indicating that wakeful conscious processing exhibits greater multiscale scale-integration than anaesthetic states (Milinkovic et al., 2025b), suggesting its utility as a dynamical and functional marker of conscious states. The sense that perceptions, thoughts, and feelings are bound into a single structured moment (Cleeremans, 2003; Searle, 2013) may depend as much on inter-scale coupling as on intra-scale interactions.

This is not an altogether new concept. Metzinger has argued that consciousness carries a substantial metabolic cost, and that any acceptable theory must show how consciousness "pays for itself" (Metzinger, 2009). Our framework offers a mechanistic interpretation: *multiscale integration is metabolically advantageous*. By tightly coupling processes across scales, the brain can reuse computational work performed on one level to constrain and guide computations at another, reducing the need for energetically expensive, scale-specific processing.

From this perspective, the brain's need to minimise metabolic expenditure whilst meeting increasingly complex behavioural and cognitive demands gives rise to the computational structuring that integrates information both horizontally across specialised functional regions, and vertically across organisational scales. Conscious experience reflects this organisation: it is the experiential manifestation of a system that, under energy scarcity, has evolved to co-determine and integrate spatially extended and temporally unfolding information into a single unified conscious experience. The characteristic unity and temporal flow of conscious experience are not accidental by-products but expressions of a multiscale integration strategy emerging from metabolic optimisation.

Energy scarcity forces the brain to coarse-grain information across scales, creating macroscopic continuous dynamics from microscopic discrete events. This coarse-graining is a physical necessity shaping computation. Conscious processing may require this metabolically driven scale-integration: the ability to simultaneously access fine-grained detail (differentiation) and coarse-grained global structure (unity) without computing each independently. Digital systems, unconstrained by metabolic scarcity, can afford to maintain strict scale-separability, but this very separability may preclude the style of processing required for consciousness.

The corollary is that consciousness is *neither* reducible to microlevel dynamics nor fully explicable by macrolevel functional patterns alone. Rather, consciousness emerges from the interdependencies and coupling across scales constrained by the physical properties of the brain (Storm et al., 2024; Aru et al., 2023; Milinkovic et al., 2025b; Dykstra et al., 2025). Simply, there is no privileged scale of computation.

In addition, the continuous physical substrate – the biological medium itself – may be essential for consciousness in ways that go beyond implementation. Continuous fields do not merely modulate information; they provide a medium for scale coupling. Discrete systems, even when hierarchically organised, lack such continuous media and therefore cannot support real-time bidirectional coupling across scales. In digital systems, information at layer k is computed, stored, and then passed to layer $k+1$ in discrete time steps. There is no continuous field that simultaneously exists across scales, enabling them to mutually constrain one another in real time. This may explain why purely discrete systems, regardless of their sophistication, cannot replicate the phenomenology of consciousness: they lack the substrate necessary for scale-integrated, hybrid computation.

Similarly the temporal continuity of conscious experience, the flowing, stream-like quality described by William James (McDermott, 2013), may depend on physical and dynamical continuity. Discrete systems update in time-steps; even if small, discontinuities remain. Biological systems, by contrast, evolve continuously in physical time, and electrophysiological and oscillatory fields provide a continuous temporal extendedness within which discrete events are embedded. This may be essential for the phenomenology of temporal flow in consciousness.

These features – hybrid dynamics, scale-inseparable organisation, metabolic embedding, and temporal continuity – may constitute necessary conditions for consciousness. They do not violate computability but redefine its scope, blending discrete and continuous modes within dynamical organisation inseparable from the biological substrate.

If we seek to build conscious machines, the implication is clear: it is not enough to scale digital systems. We must engineer systems capable of continuous-discrete interplay and multiscale inseparability. Such systems would embody metabolic constraints shaping dynamical structure, coarse-graining of stochastic microscales into computable macroscales, and field effects enabling non-local coupling unavailable in discrete systems.

Without these features, artificial systems may remain unable to replicate the computational structure underlying conscious experience. The question is not whether consciousness requires specific algorithms or functional organisations – though it may – but whether it also requires a physical substrate that natively supports the principles outlined here. If so, then silicon-based digital computers, regardless of their sophistication, may be insufficient. Instead, biomimetic substrates – neuromorphic, fluidic systems, iontronic devices, or even synthesised biological tissue – may be a good starting point to realise biological computations capable of giving rise to consciousness.

4. Implications for artificial consciousness

If consciousness relies on the hybrid, scale-integrated modes of neural computation, then current AI systems running on von Neumann architectures are unlikely to meet these conditions. Contemporary AI operates on externally powered hardware unconstrained by any intrinsic energy economy and lacks any coupling between metabolism, computation, and functional organisation. In contrast, biological systems are shaped by energy scarcity; whereby energy limits do not merely constrain computation but actively determine its form. This is not only a limitation of present-day LLMs, but also reflects gaps in theoretical frameworks for consciousness that neglect the computational implications of metabolic embedding, scale-integrated processes, and dynamico-structural co-determination.

Take, for example, recent efforts to import the Global Neuronal Workspace Theory (GNWT) (Dehaene et al., 2011, 2021) into artificial networks (VanRullen and Kanai, 2021; Goyal et al., 2021; Butlin et al., 2023). These approaches highlight both the appeal and the difficulty of translating high-level functional descriptions into artificial implementations. While they succeed in modelling aspects of access, routing, and global broadcasting, they remain committed to computational functionalism: the assumption that consciousness depends only on functional organisation, independent of the physical substrate. As we have argued, such *scale-privileging* assumptions are insufficient. Biological computationalism challenges this on two fronts: consciousness does not reside at a single privileged scale; and its computational basis is not merely functional, but intrinsically dynamico-structural, hybrid, and scale-integrated.

Even ANNs that loosely mimic some organisational motifs of the brain do not meet these conditions. As software systems, ANNs simulate functions to arbitrary precision, yet these simulations remain separable from the physical substrate that implements them. Computational time and physical time are decoupled: an implicit assumption that contrasts sharply with neural dynamics, where physical time *is* computational time (Indiveri et al., 2011). Moreover, ANNs approximate continuous functions rather than perform continuous computations natively. Whether continuous dynamics confer advantages for cognition or consciousness remains an open empirical question, but the brain's continuum-like properties suggest this is a promising line of inquiry.

It is equally important to distinguish our position from radical biological naturalism (Searle, 2017). We do not claim that only biological tissue can instantiate consciousness, nor that present-day biophysics is uniquely privileged. Whether other substrates could reproduce the requisite computational organisation, and the corresponding dynamico-structural co-dependencies, remains an open empirical and engineering question. Our aim is instead to identify which biological computational principles are necessary.

From this vantage point, we believe current debates have underappreciated the computational significance of biological organisation itself. We therefore advocate for a biology-centred conception of computation, one that takes the physical, metabolic, hybrid, and scale-integrated dynamics of neural tissue as foundational.

On the basis of these arguments, artificial systems aiming to approximate conditions relevant for consciousness would need to meet the following criteria (illustrated in Fig. 3):

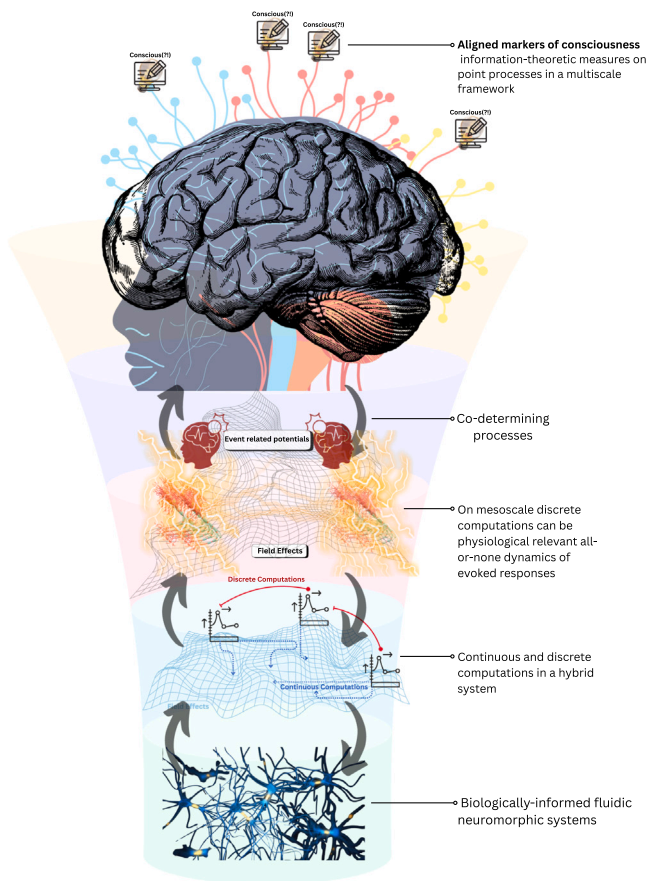

Fig. 3. A conceptual framework for scale-integrated hybrid systems. This schematic illustrates a conceptual design pathway for synthetic systems capable of approximating consciousness-relevant computation. At the base lie biologically informed substrates, such as fluidic or iontronic architectures, whose dynamics unfold in real physical time under metabolic and energetic constraints. Building upward, the figure highlights how continuous processes (field effects, graded dynamics) and discrete events (spikes, point processes) must co-determine one another within a hybrid computational architecture. At intermediate scales, biologically meaningful discrete transitions, such as evoked responses, emerge and integrate with surrounding continuous electric fields, depicted on the mesoscale of the diagram. At the macroscopic level, long-range areal interactions integrate with ubiquitous continuous field-like processes, forming system-level organisation. At the apex, aligned markers of consciousness, such as information-theoretic metrics applied to multiscale point processes, serve as indicators of emergent closure across scales. Importantly, this schematic is not a literal anatomical rendering, but a conceptual design framework: substrate-agnostic yet requiring scale inseparability, hybrid discrete–continuous dynamics, metabolic embedding, and dynamico-structural co-determination as core organisational principles.

1. *Hybrid computation*: combining continuous dynamics (e.g., electric fields, graded potentials) with discrete events, governed by *real* physical time.
2. *Scale-inseparability and metabolic embedding*: every neural process is simultaneously constrained by energetic budgets and embedded in heterarchically organised, scale-integrated processes, from molecules to whole-brain dynamics.
3. *Dynamico-structural co-determination*: brains change their own circuitry across the lifespan; substrates aiming to host consciousness must be equally dynamico-structurally adaptive.

This suggests that a serious attempt at bridging the gap to artificial consciousness requires a fundamental shift: from algorithmic abstractions to physical grounded computations where the implementational, computational and algorithmic layers are not distinct but co-instantiated. Systems capable of supporting consciousness must integrate energy, structure, and dynamics, not merely manipulate symbols.

4.1. Edge cases: Neural cultures and multiscale ANN architectures

Several emerging research directions illustrate partial expressions of these principles. One striking example comes from laboratory-grown biological neurons embedded in closed-loop learning tasks. These neural cultures can be either 2-dimensional grown on arrays or 3-dimensional neural *organoids* grown on multi-electrode arrays (*DishBrains*) which have been shown to learn control tasks with remarkable sampling efficiency: reconfiguring their functional connectivity in task-contingent ways that outperform deep reinforcement learning baselines (Khajehnejad et al., 2025). Though these are specifically tasks testing intelligence rather than sentience, this nonetheless demonstrates that biological matter, with its intrinsic dynamico-structural coupling and continuous processes, affords computational inductive biases that might be difficult to replicate in digital computers.

Further attempts have been made to bridge such neurobiological processing with formal algorithmic descriptions. Using active inference, Paul and colleagues (Paul et al., 2025) built generative models directly informed by the experimental input–output mappings of DishBrain neural cultures. By capturing how stimulation and spiking patterns drive behaviour in a simple closed-loop game, they showed that the learning dynamics of living tissue can only be reproduced *in silico* if substrate-specific constraints and priors are respected. Here, a general algorithmic principle such as predictive coding takes on novel, energy-efficient forms when adapted to biological matter.

Meanwhile, work in purely ANN software has also taken inspiration from biological multiscale organisation. The recently-proposed Hierarchical Reasoning Model (HRM) introduces two coupled timescales – a slower 'planner' and a faster 'worker' – within a small recurrent architecture trained without back-propagation through time (Wang et al., 2025). Despite its simplicity, HRM achieves near-perfect performance on Sudoku and maze tasks, and strong results on the ARC benchmark, all without pre-training or explicit chain-of-thought supervision. This new foundational model underscores how importing multiscale, biologically-inspired organisational principles yields computational capacities absent from conventional ANN designs. Yet, models like HRM still run on conventional von Neumann hardware, and it remains an open question whether such software-level design could gain further advantages if realised on synthetic, wetware-like substrates built for multiscale processing.

4.2. Neuromorphic systems: Where implementation matters, sometimes

Similar to our proposition, neuromorphic engineering has faced similar challenges regarding hardware–software distinctions (Laydevant et al., 2024). The term neuromorphic is frequently misused to describe diverse hardware and software systems that are often far from biologically realistic.

It is therefore helpful to separate neuromorphic systems into three broad categories, so that we can identify which of them are most promising for instantiating biological computationalism. First, as originally formulated by Mead (2002), neuromorphic systems referred to silicon neurons (SiNs) which were initially designed to mimic the physical operations of neuronal dynamics by explicitly implementing, not simulating, sub-threshold dynamics (Indiveri et al., 2011). In order to simulate ion channels' continuous dynamics, and in turn, their sensitivity to continuous inputs, subthreshold transistor dynamics allowed for the gradual ebb and flow of information to be implemented directly in the hardware (Mead, 2023). In these earlier neuromorphic systems, the design strategy aligned with our third criterion of dynamico-structural co-determination, whereby the algorithm is inseparable from the physical implementation. As a result, physical time and computational time were also one and the same, thereby constraining the flow of the system's information based on the substrate.

With the introduction of memristive devices, a second broad category of neuromorphic systems was introduced. Memristors – resistors which showed that the long-term memory of synapses could be implemented in solid-state electronic hardware (Strukov et al., 2008) – substantially galvanised the field of neuromorphic computing. However, many of these devices use simple weight update rules and therefore deflate the complexity of real synapses and the myriad ways plasticity and learning occur (El Boustani et al., 2012). These systems, though successful in some industrial contexts, do not follow clear neural design principles, which were central for earlier SiNs.

Third, many contemporary neuromorphic systems now consist only of software-simulated spiking neural networks running on conventional hardware – typically complementary metal–oxide–semiconductor (CMOS) processors – whose physical properties bear little resemblance to the biological substrates they are intended to emulate. The most prominent example is the SpiNNaker platform (Furber et al., 2014), frequently cited as a milestone in scaling up neuromorphic technology. Yet SpiNNaker is, at its core, a massively parallel digital simulation environment: it employs simplified neuronal models, relies on discretised time-stepping, and fully decouples computational time from physical time. These systems operate exclusively in the digital domain and do not instantiate any of the continuous, electrochemical, or energetically-embedded dynamics characteristic of biological tissue.

The success of scaling this particular class of simulated neuromorphic systems, with a candidate example in SpiNNaker (Markram et al., 2011; Furber, 2016), has sometimes encouraged the view that ever larger digital neural simulations might eventually approximate, or even emulate, biological intelligence, and by extension, cognition and consciousness. This interpretation runs the risk of conflating two very different endeavours. Large-scale digital simulations and ANN accelerators share methodological lineage and engineering goals; they solve problems through discrete symbolic operations implemented on silicon. Biological computation does not. As a result, decades of scaling simulated neuromorphic systems to millions of neurons have not yielded qualitatively new insights into brain function, nor have they demonstrated computational advantages over conventional high-performance computing and ANNs.

This is the critical point: current AI and memristive or simulated neuromorphic approaches do not share the same aims as the earlier biologically grounded neuromorphic systems. If we are to build synthetic systems that fulfil the tripartite criteria we outlined, neuromorphic technologies must contribute something fundamentally new. Their success will not come from reproducing the architecture or engineering logic at the scale of today's AI models, but from embracing a radically different computational paradigm: one rooted in physical substrates that implement the same dynamico-structural principles found in biological brains (Indiveri et al., 2011; Laydevant et al., 2024; Indiveri, 2025).

Out of the three broad categories outlined, it is evident that only the earlier biologically faithful neuromorphic hardware systems (SiNs) converge on the same goal we offer here: computation is not independent of its substrate, and energy efficiency is not the goal but rather the constraint that induces a multiscale, scale-integrated computational regime.

4.3. Towards new substrates

BrainScaleS-2 (BSS-2) represents a major advance over purely digital neuromorphic platforms like SpiNNaker (Pehle et al., 2022). Unlike SpiNNaker – which runs discretised simulations – BSS-2 uses mixed-signal analogue circuits whose membrane and synaptic dynamics unfold through the physical behaviour of capacitors and conductances. This means that, unlike digital systems, BSS-2 implements synthetic neural dynamics directly in continuous time, partially collapsing the algorithm-implementation divide. However, key divergences remain. BSS-2 operates in an accelerated physical regime, roughly three orders of magnitude faster than biological time (Pehle et al., 2022), and it is still realised in rigid silicon hardware. As such, it lacks crucial biological features such as ionic gradients, diffusion-coupled signalling, and structured stochasticity. For these reasons, BSS-2 narrows the gap between digital simulation and biological computation, but it does not yet meet the tripartite criteria our framework identifies as necessary for consciousness-supporting computation.

If synthetic systems are to approximate the computational regime outlined in this paper, emerging hardware improvements moving beyond simulated spiking networks on CMOS, and beyond solid-state memristors, towards systems where the substrate physics instantiate the computation need to be considered.

A review by Law et al. (2025) highlights how new fluid-based neuromorphic devices implement computations through the motion of ions, diffusive coupling and electro-osmotic transport, coming much closer to our original sentiment. Aligning with biological computationalism, these fluidic devices operate in true physical time, with state evolution governed by chemical gradients, diffusion dynamics and nonlinear ionic transport. For this reason, these systems sit nearer to the biologically-relevant end of our neuromorphic systems: they are hardware instantiations of neural design, not mere simulations; they also exhibit continuous, substrate-dependent dynamics; and they provide a medium where computation and dynamics are not separable.

Two landmark contributions by Xiong and colleagues (Xiong et al., 2023b,a) have furnished this shift. They demonstrate a fluidic memristor, where computations arise from the history-dependent reorganisation of ions within a charged microchannel. These devices realise neural-like dynamics, particularly nonlinear synaptic dynamics emerging directly from the physics of the ion transport rather than numerical simulations. The authors further argue (Xiong et al., 2023b) that bringing in chemistry yields much richer internal state spaces and structured stochasticity, which is far more emblematic of biological synapses (Destexhe and Rudolph-Lilith, 2012), and is not possible in solid-state memristors. This recent advance offers compelling evidence that fluidic memristive neuromorphic systems realise substrate-embedded activity with biological timescales of computations and stochasticity built in.

As it stands, these developments suggest that the future of neuromorphic engineering may not lie in scaling up digital simulations of spiking networks, but in expanding technologies implementing physically grounded computational media. Fluidic devices offer chemical richness, structured neural noise, and graded potentials: candidate properties tied to the tripartite criteria we outlined for biological computationalism. Although it remains to be seen how this might be practically applied, given that none of these recent advancements currently realise fully multiscale heterarchical biological computations, they nevertheless demonstrate a decisive shift away from disembodied computations of the von Neumann kind. If artificial consciousness requires a computational ontology grounded in the physics of the substrate, as we argue, it is these fluidic, biologically-plausible neuromorphic hardware systems, rather than scaled-up digital simulations, that offer the most promising foundations for the next generation of biologically-informed synthetic minds.

4.4. Building aligned markers of consciousness

If artificial consciousness is to be meaningfully evaluated, then our measures of consciousness must also reflect the computational principles we claim consciousness depends on. As system-agnostic measures that do not depend on the mechanism, neuroscience has relied on information-theoretic and complexity-based biomarkers including, but not limited to, PCI (Casali et al., 2013), Lempel–Ziv complexity (Schartner et al., 2015), neural complexity (Tononi et al., 1994), integrated information (Balduzzi and Tononi, 2008), and related dynamical metrics. These are simple-to-use operational indices of conscious level. They mostly quantify algorithmic or dynamical complexity either at one level of description or over signals pre-processed into a single temporal structure. As a result, the standard toolbox cannot capture the multiscale emergent structure realised in biological systems. As an advancement on this challenge, measures of emergence that either capture synergistic activity (Mediano et al., 2025) or the closure of the macroscopic process from the microscopic (Barnett and Seth, 2023; Milinkovic et al., 2025a,b), have been used to evince a multiscale approach. Extending these frameworks would be central to developing simple, measurable assessments of conscious state based on our criteria.

Consequently, it is paramount to extend our mathematical and information-theoretic formalisms to match the computational ontology of biological tissue. What is required is a theory of multiscale stochastic processes capable of modelling systems where discrete events are embedded within, and shaped by, continuous dynamics. Here, point process theory provides one pathway (Jacobsen, 2006): it naturally describes systems in which events are generated and embedded in continuous processes, and where discrete and continuous influences co-determine the system's dynamics.

Foundational progress has already been made at the single scale. Spinney et al. (2017), Spinney and Lizier (2018) and colleagues (Shorten et al., 2021) have developed rigorous formulations of information flow and information storage for continuous-time, continuous-valued multivariate stochastic processes. These methods allow the characteristic information dynamics to be quantified in systems where dynamics evolve continuously in physical time, i.e., precisely the regime relevant for neurobiology and for physically instantiated hybrid systems.

The challenge, and the opportunity, is to generalise these advances to emergent multiscale frameworks such as those used to study scale-integration across anaesthetic states and wakefulness (Milinkovic et al., 2025b). Conscious processing, on our account, is not merely information exchange within a single dynamical level; it is information flow across levels, where coarse-grained macroscale processes constrain and are constrained by fine-grained microscale processes in real time. Measures of information dynamics must therefore track not only how information is stored or transferred within a scale, but how it is transformed, compressed, dissipated, or reused across scales. A multiscale, continuous-time information-theoretic framework, one that unifies point-process formalisms with stochastic differential models and cross-scale operators, is thus an exciting avenue to pursue to operationalise biological computationalism.

Such measures would serve two essential roles: first, they would provide biologically aligned biomarkers – indices sensitive to hybrid, scale-integrated dynamics – and second, they would function as benchmarks for synthetic systems, allowing us to determine whether an artificial substrate reproduces the multiscale information-processing patterns characteristic of conscious biological systems. If consciousness depends on the biological computations as we have outlined, then the operational measures of consciousness must be adapted accordingly. Only with such multiscale, hybrid-sensitive measures can we meaningfully assess whether a synthetic system approximates the informational signatures of consciousness or merely simulates them. We offer a schematic flow of the proposed implementation in Fig. 3.

5. Conclusion

The ideas outlined here define the contours of a biological theory of computation that would aid in understanding consciousness in artificial systems and other things: one grounded in cross-scale coupling, energy, and continuous-discrete dynamics, and capable of bridging the emergent multiple scales of neural processes, from molecular mechanisms to conscious processing (Sacha et al., 2025). Developing such a framework is not only a theoretical challenge but also an empirical one demanding integration of synthetic engineering, neurophysiology, computation in dynamical systems, and information-theoretic frameworks into a unified account of computation in evolved systems.

In this sense, the future of artificial consciousness research may lie not in scaling up existing models but in embracing a paradigm of biosynthetic, energy-aware, hybrid computation which realise emergent scale-integrated dynamical structure. This requires moving towards a genuine *biological computationalism*.

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